Tax Productivity and Economic Development: A Quantitative Macroeconomic Analysis

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Tax Collection and Its Role in Development

- Ability to raise taxes central to state capacity theory of development

  Tilly (1990), Besley and Persson (2011), Besley, Ilzetzki, and Persson (2013), Mayshar, Moav, and Pascali (2022), D'Arcy et al. (2024)

- Adam Smith in *The Wealth of Nations*:

  "Every tax ought to be so contrived as both to take out and to keep out of the pockets of the people as little as possible over and above what it brings into the public treasury of the state. . . the cost of collecting it may be very considerable, and in the worst cases may amount to more than it brings in."

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- Yet largely absent from macro development: models of public capital assume *costless* taxation

  Barro (1990), Baxter and King (1993), Glomm and Ravikumar (1992), Ramey (2021)

- Little consensus on *quantitative importance* of tax collection capacity

This Paper

1. **New data** on tax collections (T) and tax administration expenditures (X)
   - Hand-collected from national authorities: 98 countries + 16-country historical panel + US and Indian states
   - **Fact**: T/X rises sharply with development: $\approx \$30$ per \$1 spent in poorest countries $\rightarrow$ $\approx \$270$ in richest

2. **Model**: neoclassical growth model + tax administration sector
   - Labor ("tax collectors") raises revenue $\Rightarrow$ funds public capital + transfers
   - Countries differ in productivity A_M *and* tax productivity A_C

3. **Quantify**: measure A_C in 98 countries; validate it; simulate long-run effects of raising it; explain why they are so large
   - Tax administration employs $< 1\%$ of workers everywhere... can A_C matter for development?

Preview of Main Results

1. Tax productivity varies much more with development than aggregate productivity
   - Slope of $\log A_C$ in log GDP p.c.: 0.80 vs. A_M: 0.34, PWT TFP: 0.27
   - A_C lines up with what tax authorities actually *do*; A_M does not

2. Raising A_C by 50% in low-income countries: **GDP $+18\%$, consumption $+18\%$**
   - Hulten's theorem predicts 0.01–0.4%; public capital ($+80\%$) complements private capital ($+18\%$)
   - Why: public capital is unpriced and financed by a distortionary tax, so the economy is far from efficient

3. Gains fall with development, and hiring more collectors is no substitute
   - Middle income: GDP $+15\%$, consumption $+10\%$; high income: GDP $+9\%$, consumption $+0.7\%$
   - Re-sizing the tax authority optimally adds $< 1\%$ of GDP, and a Ramsey government leaves poor countries' public capital unchanged: the marginal collector barely covers their own wage

Empirics: Tax Collections Relative to Expenditures

Measuring Tax Collections per Dollar of Expenditure

- Unit of observation: the **national tax authority**
- T = actual tax collections
  - PIT, CIT, VAT/sales, excises + customs
- X = tax authority expenditure
  - Operating + capital costs; salaries and ICT largest
- T/X = average collection per \$1 of administration
  - Unit-free; comparable across countries and time
  - *Average* (not marginal) product; model links the two





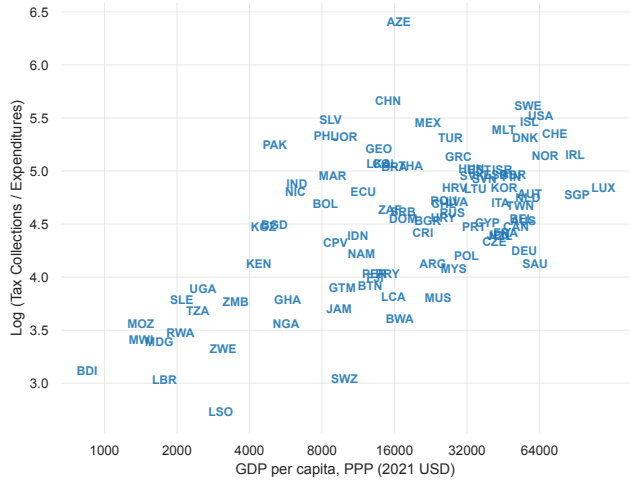
Data: Three Levels of Variation

1. **Cross-section**: 98 countries, 2014–2017
   - ISORA survey (IMF/OECD/CIAT) + national budgets and tax authority reports
   - Covers 77% of world population, 67% of world GDP

2. **Historical panel**: 16 countries, 55 years per country on average
   - US back to 1877 (IRS + customs annual reports); digitized archival records
   - Australia, Bangladesh, Botswana, Canada, Denmark, India, Kenya, Malawi, Pakistan, S. Korea, Sierra Leone, South Africa, Thailand, UK, Uruguay

3. **Sub-national panel**: sales taxes devolved to states
   - 50 US states (1993–2017) and 30 Indian states (1950–2017)

Tax Collections / Expenditure Rise Strongly with GDP per Capita



Cross-Section Regressions US Long Run Long-Run Panel: Countries and States

Robust Across Countries; Suggestive Within Countries over Time

- **Cross-section**: elasticity of T/X w.r.t. GDP p.c. $= 0.34$ (s.e. 0.05)
  - Steeper among low/lower-middle-income countries: 0.77 vs. 0.17 for upper-middle/high
  - Robust to authority scope: statutory rates, VAT, customs, social security, registry Table

- **Within countries, long run**: split each series into two ≈27-year segments, relate the long-run change in $\log T/X$ to the long-run change in $\log$ GDP p.c.
  - 16-country panel: 0.57 (s.e. 0.22); all time-invariant country characteristics differenced out
  - Robust to dropping authorities without customs (0.61) or with social security (0.57) Table
  - Same design within states: US 1.36 (s.e. 0.49); India 0.61 (s.e. 0.21)

- **Annual panel** with country (state) and year fixed effects: positive but smaller, significant only for US states

$\Rightarrow$ Robust pattern across countries today; suggestive pattern within countries over the long run

Model of Tax Productivity and Development

Model Overview: Neoclassical Growth + Tax Administration

- Neoclassical growth model where tax revenue finances **productive public capital**
  Barro (1990), Baxter and King (1993), Ramey (2021)

- Novelty: a **tax administration sector** — raising revenue requires labor
  1. Effective tax rate *produced* by tax collectors: $\tau = A_C \, L_C^{\eta}$
  2. Decreasing returns to collection ($\eta < 1$): easy tax handles first, harder ones later
     trade taxes $\rightarrow$ income and consumption taxes ... Jensen (2022)
  3. Waste: fraction $1 - \theta$ of public investment lost; share ξ of revenue to transfers

- Countries differ in $\left(A_M,\ A_C,\ \theta,\ \xi,\ L_C\right)$; all other parameters common

Households and Firms

- Representative household:

$$\sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \alpha \frac{L_t^{1+1/\varepsilon}}{1+1/\varepsilon} \right], \qquad C_t + K_{t+1} = (1 + r_t - \delta)K_t + (1 - \tau_t) w_t L_t + TR_t$$

  - Takes tax rate τ_t and transfers TR_t as given; standard consumption/savings/labor choices

- Final good (consumption or investment), competitive firms:

$$Y_t = A_M \left(\theta K_{G,t}\right)^{\zeta} K_t^{\mu} L_{M,t}^{1-\mu}$$

  - Constant returns in private inputs (K, L_M); $\zeta > 0$: increasing returns overall
  - θ: efficiency of turning public investment into effective public capital

Tax Administration and the Government

- **Tax collection technology**:

$$\tau_t \;=\; A_C \left(L_{C,t}\right)^{\eta}, \qquad 0 < \eta \leq 1$$

  - L_C: tax collectors; $\eta < 1$: each additional collector less effective than the last
  - Timing: government commits to $\{L_{C,t}\}$, households respond (Stackelberg)
  - Baseline: L_C set to match data; chosen optimally in extensions

- **Government budget**: revenues fund transfers, collectors' pay, public investment

$$\underbrace{TR_t}_{=\xi T_t} + \underbrace{X_t}_{=w_t L_{C,t}} + K_{G,t+1} \;=\; \underbrace{T_t}_{=\tau_t w_t (L_{M,t}+L_{C,t})} + (1-\delta_G) K_{G,t}$$

What T/X Identifies in the Model

- In equilibrium, wages cancel in the ratio of collections to expenditures:

$$\frac{T_t}{X_t} = \frac{A_C \, L_{C,t}^{\eta} \left(L_{M,t} + L_{C,t}\right)}{L_{C,t}}$$

- Direct link between the data statistic and tax productivity A_C
  - Given L_C and L_M: higher $T/X \Leftrightarrow$ higher A_C
  - T/X is the *average* product of collection; marginal product $\approx \eta\times$ average
- Steady state: two equations in (L_M, K) pin down full allocation Steady-State System
  - Unique if $\gamma > 1$ and $\mu + \zeta < 1$; otherwise the public-capital feedback can support multiple steady states Uniqueness

Taking the Model to the Data

General Parameters (Common to All Countries)

Description	Parameter	Value
Discount factor	β	0.96
Risk aversion / inverse IES	γ	1.1
Frisch elasticity	ε	0.5
Labor disutility weight	α	1
Capital share	μ	0.33
Depreciation (private, public)	$\delta,\ \delta_G$	0.05
Returns to scale in tax collection	η	0.05
Output elasticity of public capital	ζ	0.20

- Standard choices where available (ε: Kleven et al. 2025; γ: Bick et al. 2022; μ: Gollin 2002)
- η and ζ are less standard and quantitatively crucial $\Rightarrow$ next two slides

Returns to Scale in Tax Collection: $\eta = 0.05$

- In the model, $\eta \approx \dfrac{\textbf{marginal} \text{ product of collection spending}}{\textbf{average} \text{ product of collection spending}}$

- Micro estimates of marginal returns vs. our average T/X, across income levels:

	Evidence	Marginal	Average	Implied η
Ghana	Property tax RCT (Dzansi et al. 2025)	1.6	43	0.04
Costa Rica	Enforcement experiment (Brockmeyer et al. 2019)	4	82	0.05
United States	Random audits (Boning et al. 2025)	13	260	0.05

- Remarkably similar ratios in low-, middle-, and high-income settings

Public Capital Parameters: $\zeta = 0.20$ and θ by Income Group

- ζ (output elasticity of public capital): wide range in the literature, no consensus
  - Aschauer (1989): 0.39; Ramey (2021): 0.05 for transportation infrastructure; Cubas (2020): 0.09
  - Bom and Ligthart (2014) meta-analysis: mostly advanced economies Histogram
  - Newer evidence: US railroads ≈ 0.6 (Donaldson and Hornbeck 2016); China's highways ≈ 0 for the periphery (Faber 2014)
  - Benchmark $\zeta = 0.20$; sensitivity to 0.10 and 0.30 below

- θ (public investment efficiency): World Bank Infrastructure Efficiency Index
  - $\theta_{low} = 0.58$, $\theta_{middle} = 0.74$, $\theta_{high} = 0.84$ (Devadas and Pennings 2018)
  - Captures higher waste in public expenditure at lower incomes

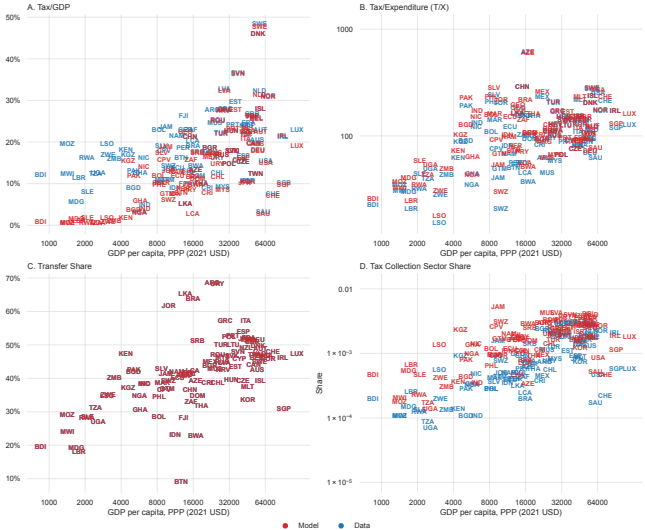
Country-by-Country Calibration: 98 Countries

Four country-specific parameters chosen **jointly** to match four moments per country:

Country-specific parameters	Targeted moments
Aggregate productivity A_M	GDP
Tax productivity A_C	Collections per expenditure (T/X)
Transfer share ξ	Transfer share of public expenditure
Tax collectors L_C	Tax authority employment share

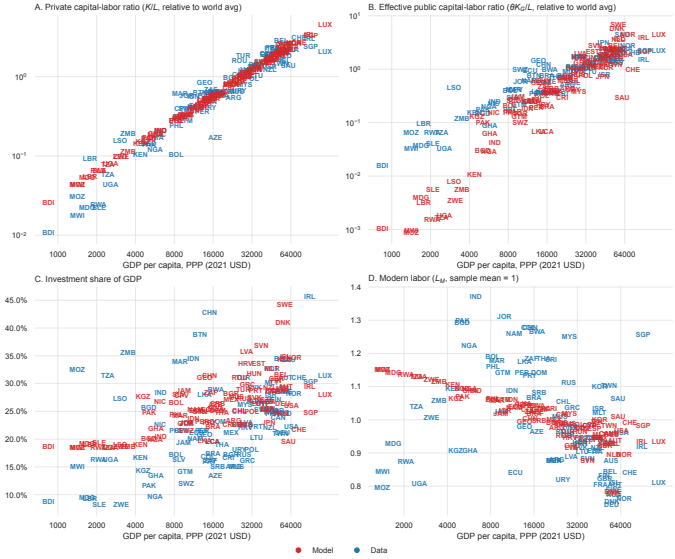
- Data: own collection (T/X, employment), IMF GFS (transfers), PWT 11 (GDP p.c.)
- The tax-to-GDP ratio is targeted jointly with T/X; results hold when either is targeted alone Robustness

Model Fit: Targeted Moments

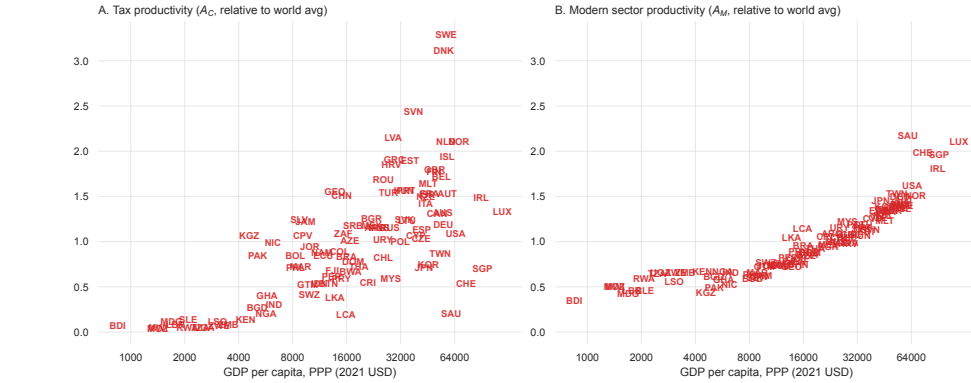


Targeting T/X or Tax-to-GDP Alone

Model Fit: Untargeted Moments



Tax Productivity Varies Far More with Development than TFP



- Slope of log productivity on log GDP p.c.: A_C : 0.80 (s.e. 0.07) vs. A_M : 0.34 (s.e. 0.015) [PWT TFP: 0.27]
- Tax collection especially unproductive in poor countries — a direct proxy for state capacity

Robustness: Collection Capital, Wage Premia, η, ζ

What Is A_C Picking Up? What Tax Authorities Actually Do

- ISORA also records the *real activities* of national tax authorities (97 of our countries): third-party data and digitization, enforcement, operational processes

Pre-filing of returns: 1.02^{***}; artificial intelligence: 0.79^{***}; garnishment: 1.04^{***} All Activities

	Outcome: A_C		Outcome: A_M
Index of tax-authority activities	No controls	$+ D_i$	$+ D_i$
Third-party data and digitization	0.579***	0.214	
	(0.118)	(0.130)	
Enforcement	0.390***	0.221**	
	(0.104)	(0.103)	
Operational processes	0.421***	0.207*	
	(0.111)	(0.105)	
Combined index	0.716***	0.348**	-0.025
	(0.168)	(0.161)	(0.128)
Observations	97	69	69

D_i: property rights, trust, digital penetration, management, schooling, trade openness, employee and agricultural shares, private credit. Controlling directly for A_M: combined index 0.535^{***} (0.171). Robustness

$\Rightarrow$ A_C is a distinct measure of tax-collection productivity — and one that policy can move

Is the A_C Gradient an Artifact? Robustness

Specification	Slope of $\log A_C$ on log GDP p.c.
Baseline: labor-only collection, collectors paid the private wage	**0.80**
Capital in collection, calibrated to non-wage share of spending	0.65
Capital in collection, calibrated to ICT share of spending	0.74
Public-sector wage premia: collectors paid $\pi_i \, w_M$	0.72
Returns to collection: $\eta = 0.10$ / 0.20 / 1	0.77 / 0.73 / 0.37
Output elasticity of public capital: $\zeta = 0.10$ / 0.30	0.80 / 0.80

- **Capital**: $\tau = A_C K_C^{\eta\psi} L_C^{\eta(1-\psi)}$; non-wage items are 27–33% of operating spending Capital
- **Wage premia** are larger in poor countries; they account for about a tenth of the gradient Wage Premia
- **Constant returns**, $\eta = 1$: rich countries' larger tax authorities explain more of their higher tax rates, but the slope (0.37) still exceeds A_M's (0.34)
- ζ does not matter: A_C is identified from T/X and L_C/L, not from the production function

Counterfactual: Long-Run Effects of Raising Tax Productivity

Experiment: Raise A_C by 50 Percent, Country by Country

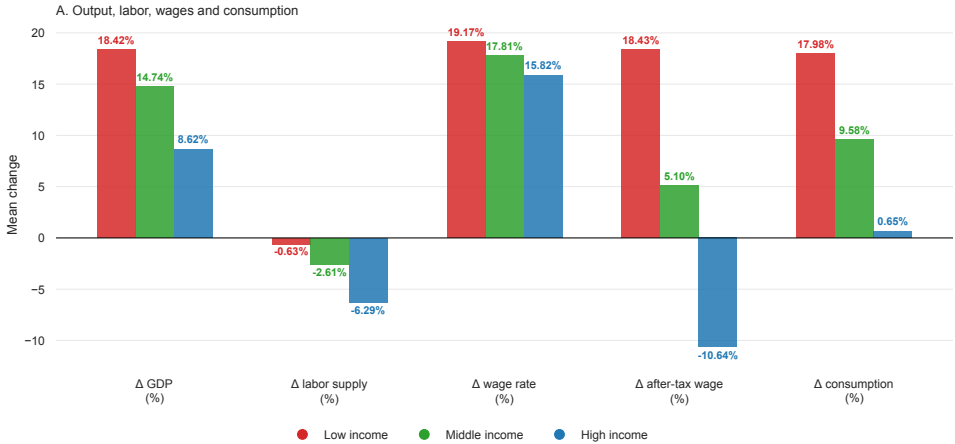
- Permanent increase in A_C of 50%; compare steady states (L_C fixed at baseline)

- A transformative but *feasible* change with existing policy levers:
  - GIS databases + tablets for property tax collectors roughly *doubled* collections in Ghana (Dzansi et al. 2025)

    See also Jawara, Knebelmann, Levine, Ndiaye, and Pouliquen (2025) — in the field, The Gambia

- Report unweighted averages by World Bank income group: low (10 countries), middle, high

Poor Countries Gain the Most; Rich Countries' Consumption Barely Moves



Steady-state changes from $A_C \times 1.5$, unweighted means within World Bank income groups

Full Figure: Capital Stocks and Tax Rates

Why Do Poor Countries Gain the Most?

With L_C fixed, $\bar{\tau} = A_C \bar{L}_C^{\eta}$ rises by 50% in every country. But starting points differ:

	Low income	Middle income	High income
Tax rate: pre $\to$ post (%)	$1.2 \to 1.8$	$16.9 \to 25.3$	$29.9 \to 44.8$
Public capital (%)	$+80.0$	$+73.2$	$+63.8$
Wage rate (%)	$+19.2$	$+17.8$	$+15.8$
After-tax wage (%)	$\mathbf{+18.4}$	$\mathbf{+5.1}$	$\mathbf{-10.6}$
Labor supply (%)	-0.6	-2.6	-6.3
GDP (%)	$+18.4$	$+14.7$	$+8.6$
Consumption (%)	$\mathbf{+18.0}$	$\mathbf{+9.6}$	$\mathbf{+0.7}$

- Poor countries: tax burden stays tiny, public + private capital expand $\Rightarrow$ workers gain
- Rich countries: $\tau \approx 45\%$ erodes after-tax wages (in practice: would shed collectors instead — see extensions)

Gains Are Orders of Magnitude Beyond Hulten's Approximation

GDP effect of $A_C \times 1.5$, low-income countries	Percent change
Model	**18.4%**
Hulten's prediction (wage-bill share: $0.5 \times wL_C/Y$)	0.01%
Hulten's prediction (tax share: $0.5 \times T/Y$)	0.41%
Direct public capital channel ($\zeta \times \%\Delta K_G$)	16.0%

- Hulten: aggregate gain $\approx$ sectoral TFP gain $\times$ Domar weight $\Rightarrow$ tiny for tax administration
- Why it fails here: the tax authority sells nothing, and public capital is unpriced and financed by a distortionary tax — the allocation is not efficient, so reallocation has first-order effects
- Most of the gain: public capital channel; remainder from induced private capital

Where Do the Gains Come From? The Public-Capital Loop

- Simplest case: static, no private capital, inelastic labor, all revenue invested:

$$Y = A_M(\theta K_G)^{\zeta} L_M, \quad \tau = A_C L_C^{\eta}, \quad K_G = \tau Y \quad \Longrightarrow \quad \frac{d\log Y}{d\log A_C} = \frac{\zeta}{1-\zeta} = 0.25$$

- More revenue buys public capital, which raises output, which raises revenue at the same tax rate: $\zeta + \zeta^2 + \zeta^3 + \cdots = \zeta/(1-\zeta)$ (Barro 1990)

- Hulten instead assigns the Domar weight $\lambda_C = wL_C/Y$: ≈ 0.0002 in low-income countries
  - Hulten is an envelope result at an efficient allocation (Baqaee and Farhi 2019, 2020); here K_G is unpriced and τ is a wedge

- Full dynamic model: private capital joins the loop (Baqaee and Malmberg 2025)

$$\frac{d\log Y}{d\log A_C} = \frac{\zeta\,\Psi}{1-\mu-\zeta} \approx 0.43, \qquad \Psi = 1 + \frac{\lambda_C}{s_G} \leq 1.03, \quad s_G = \delta_G K_G/Y$$

  - $1.5^{0.43} - 1 \approx 19\%$, close to the model's 18.4%; the Domar weight enters only as a small correction inside Ψ

Why Do Gains Fall with Development? Labor Supply and Transfers

- Add elastic labor and a transfer share ξ (static, no private capital):

$$\frac{d\log Y}{d\log A_C} = \underbrace{\frac{\zeta}{1-\zeta}}_{\text{loop}} + \frac{1}{1-\zeta}\cdot\frac{\mathcal{I}-\mathcal{S}+\mathcal{W}}{D}$$

  - $\mathcal{S} = \tau/(1-\tau)$: **substitution** — lower after-tax wage, work less
  - $\mathcal{I} = \gamma\,\varphi/(1-\varphi)$, $\varphi = (1-\xi)\tau$: **income** effect of paying for public capital — work more
  - $\mathcal{W} = (1-\gamma)\,\zeta/(1-\zeta)$: **wealth** effect of higher output — take some of it as leisure
  - $D = s/\varepsilon + (\gamma-\zeta)/(1-\zeta)$, $s = L_M/L$: the smaller the Frisch elasticity, the smaller every response

- Without transfers, $\mathcal{I} = \gamma\mathcal{S}$ more than offsets $\mathcal{S}$. Transfers shrink $\mathcal{I}$ but not $\mathcal{S}$, and ξ rises with income:

	Low income	Middle income	High income
$d\log Y/d\log A_C$, no transfers ($\xi = 0$)	0.241	0.252	0.260
$\mathcal{I}-\mathcal{S}$, with transfers	-0.01	-0.13	-0.28
$d\log Y/d\log A_C$, with transfers	**0.24**	**0.19**	**0.13**

Does Choosing L_C Optimally Undo This? Only If η Is Large

- Government chooses L_C to maximize welfare; output responds directly and through L_C:

$$\frac{d\log Y}{d\log A_C} \;=\; \Lambda \;+\; (\eta\Lambda - \kappa)\,\frac{d\log L_C}{d\log A_C}$$

  - Λ: elasticity with L_C fixed; $\eta\Lambda - \kappa$: a collector raises τ by $\eta\%$ but pulls labor out of production

$d\log Y/d\log A_C$	$\eta = 0.05$	$\eta = 0.20$	$\eta = 0.50$	$\eta = 1$
Low income	**0.236**	0.228	0.112	0.008
Middle income	**0.166**	0.071	0.014	0.004
High income	**0.083**	0.029	0.009	0.003

- Small η: τ pinned by A_C, payroll barely moves $\Rightarrow$ same as with L_C fixed (0.236 vs. 0.237)
- $\eta = 1$: A_C and L_C are perfect substitutes; the government keeps τ and shrinks its payroll $\Rightarrow$ **Hulten is restored** (marginal = average product of collection)

$\Rightarrow$ Gains beyond Hulten survive because returns to collection are sharply decreasing Inelastic Labor Full Model

Alternative Policies for Low-Income Countries

New steady state relative to baseline	$A_C \times 1.5$	$\theta \to 0.84$	$L_C \times 1.5$	$A_M \times 1.5$
GDP	1.18	1.16	1.00	2.28
Private capital	1.18	1.16	1.00	2.28
Public capital	1.80	1.16	1.00	2.28
Consumption	1.18	1.16	1.00	2.28
Transfers	1.78	1.16	1.02	2.28
After-tax wage	1.18	1.17	1.00	2.34
Labor supply	0.99	0.99	1.00	0.97
T/X ratio	1.49	1.00	0.68	0.97
Tax rate	1.50	1.00	1.02	1.00

- Cutting waste (θ: $0.58 \to 0.84$, rich-country level) $\approx$ as powerful as raising A_C
- **Hiring 50% more collectors achieves nothing**: strong decreasing returns ($\eta = 0.05$), and T/X falls by a third
- $A_M \times 1.5$: huge gains, but no known policy lever of that size

Is Public Capital Under-Provided? A Planner Benchmark

- Planner allocates inputs directly (no collection costs):

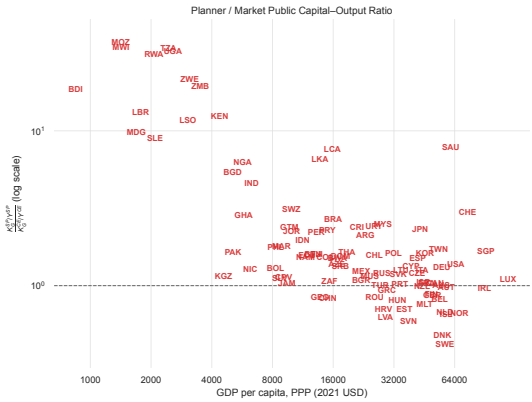
$$\frac{K_G^{SP}}{Y^{SP}} = \frac{\zeta}{\beta^{-1} - 1 + \delta_G} \approx 2 \quad \forall \text{ countries}$$

- Market outcome depends on τ, ξ, L_C — *not* on ζ:

$$\frac{K_G^{CE}}{Y^{CE}} = \frac{1-\mu}{\delta_G} \cdot \frac{(1-\xi)\tau(L_M + L_C) - L_C}{L_M}$$

- Low-income countries: public capital **an order of magnitude below** the efficient level (≈ 10–$40\times$)

- Middle income: mostly within 1–$3\times$; high income: around one



Efficient / market public capital-output ratio (log scale)

Sensitivity: ζ Scales the Gains; under Constant Returns They Vanish

		Public capital (ζ)		Collection (η)		
	Baseline	0.10	0.30	0.10	0.20	1
Cross-country productivity slopes						
Slope of $\log A_M$ in log GDP p.c.	**0.34**	0.53	0.16	0.34	0.34	0.34
Slope of $\log A_C$ in log GDP p.c.	**0.80**	0.80	0.80	0.77	0.73	0.37
$A_C \times 1.5$, *low-income countries*						
Δ GDP (%)	**18.8**	7.4	38.3	19.8	22.3	1.1
Δ Consumption (%)	**18.3**	6.9	37.8	19.3	21.6	0.6
$A_M \times 1.5$, *low-income countries*						
Δ GDP (%)	**127.7**	98.2	180.7	127.6	127.2	127.7

Each column recalibrates the model; L_C is chosen optimally before and after each shock.

- Gains scale with ζ (the loop multiplier $\zeta/(1-\mu-\zeta)$); the A_C gradient does not move
- $\eta = 1$: A_C and collectors are perfect substitutes, so the government takes the gain as a smaller payroll $\Rightarrow$ close to Hulten; the A_M shock is unaffected
- Micro evidence puts marginal / average collections at 0.04–0.05: η is far from one

Extensions

Extensions: Relaxing Four Baseline Assumptions

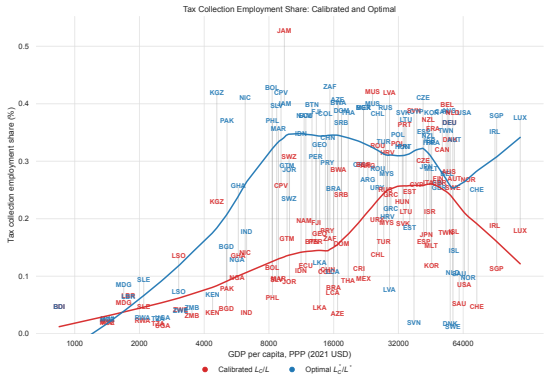
Baseline assumption	Extension
Size of the tax authority is calibrated	I. Government chooses L_C II. Ramsey taxation with borrowing
All output is taxable and uses public capital	III. Untaxed traditional sector
Collection uses only labor	Capital in tax collection Slide
Collectors earn the private-sector wage	Public-sector wage premia Slide

- The last two change how A_C is measured: gradient $0.80 \rightarrow 0.65$–0.74 (capital), 0.72 (wage premia)

Extension I: Letting the Government Choose L_C

- Choose L_C to maximize steady-state household welfare, all else at calibrated values
- **Low- and middle-income countries employ too few collectors**: calibrated $L_C \approx 78\%$ and 53% of the optimum
- High-income countries: above the optimum on average (mean ratio 2.4)
- But gains from re-optimizing L_C are small: $\Delta C \approx 0.1$–0.8% Table

$\Rightarrow$ Low A_C, not misallocated L_C, is the binding constraint



Extension II: Ramsey Taxation with Costly Collection

- Government chooses $\{L_{C,t}\}$ to maximize lifetime welfare; labor supply fixed. It can borrow and tax capital, but public investment, transfers, and collection are paid out of collected taxes Problem

- **Proposition**: at an interior steady state, public capital satisfies

$$\underbrace{F_{K_G} + \frac{\Phi\, F_{L_M K_G}}{\mathcal{M}}}_{\text{social marginal product}} = \underbrace{\left(\beta^{-1} - 1 + \delta_G\right)}_{\text{user cost}} \underbrace{\left(1 + \frac{1}{\mathcal{M}}\right)}_{\text{collection markup}}, \qquad \mathcal{M} = (1-\xi)\,\eta\,\frac{T}{X} - 1 + \frac{\mu\, s_G}{1-\mu}$$

- $\mathcal{M}$: what the marginal collector adds to public investment, net of their own wage $\Rightarrow 1/\mathcal{M}$ is a marginal cost of public funds from collection alone (Keen and Slemrod 2017)

- Costless collection: $\mathcal{M} \to \infty \Rightarrow$ planner's benchmark. Costly collection:

$$\left.\frac{K_G}{Y}\right|_{\text{Ramsey}} = \frac{\zeta}{\beta^{-1} - 1 + \delta_G} \cdot \frac{\mathcal{M}}{\mathcal{M} + 1 - s_G^{SP}}, \qquad s_G^{SP} \approx 0.11$$

- With $\eta = 0.05$, doubling the tax authority nearly halves $\eta\, T/X$: hiring raises the markup much faster than revenue, while higher A_C raises $\mathcal{M}$ at every size

Ramsey Steady States: Hiring Cannot Close the Gap in Poor Countries

	Low income	Middle income	High income
Baseline calibration (group averages)			
Tax rate τ (%)	1.22	16.86	29.86
Collection employment L_C/L (%)	0.030	0.160	0.250
Net marginal revenue $\mathcal{M}$	0.51	2.11	2.11
K_G/Y relative to efficient	0.05	0.60	0.93
Ramsey steady state			
Tax rate τ (%)	1.24	17.11	28.39
Collection employment L_C/L (%)	0.044	0.216	0.091
Collection markup $1 + 1/\mathcal{M}$	20.6	1.74	1.14
K_G/Y relative to efficient	0.05	0.60	0.89

- Low income: hires $\approx 50\%$ more collectors, yet K_G/Y stays at 5% of efficient — the marginal collector barely covers their own wage
- Micro evidence agrees: a marginal return of 1.6 in Ghana implies $\mathcal{M} \approx 0.2$, vs. ≈ 5.7 for 13 in the US
- High income: cuts collectors by about two-thirds; K_G/Y at 89% of efficient

Extension III: Untaxed Traditional Sector

- Add linear, untaxed traditional sector: $Y_T = A_T L_T$; no benefit from public capital
  - A_T calibrated to each country's agricultural employment share Calibration

- Raising A_C by 50% in low-income countries: **GDP** $+6.2\%$**, consumption** $+5.0\%$
  - Public capital still doubles; smaller aggregate gains since traditional sector unaffected
  - Still far above Hulten: 0.01% (wage-bill share), 0.31% (tax share) Counterfactuals Hulten

- Targeting *unmechanized* agriculture instead: GDP $+13\%$, consumption $+11\%$ Robustness
  - Rich-country agriculture is mechanized and taxed — poor proxy for the untaxed sector

Conclusion

- New data on tax collections per dollar of administrative expenditure: rises from $\approx \$30$ to $\approx \$270$ with development — robustly across countries, suggestively within countries over the long run
- Through the lens of the model: tax productivity varies **far more** with development than aggregate TFP (slope 0.80 vs. 0.27–0.34)
- A_C tracks what tax authorities actually do (third-party data, digitization, enforcement); A_M does not $\Rightarrow$ tax productivity is distinct, and movable by policy
- Raising tax productivity 50% in low-income countries: **GDP** $+18\%$**, consumption** $+18\%$ — orders of magnitude beyond Hulten's approximation
- Source: a public-capital loop that Hulten's theorem misses because public capital is unpriced; it survives optimal hiring of collectors because returns to collection are sharply decreasing
- Public capital in poorest countries far below efficient levels; even a Ramsey government cannot close the gap by hiring — collecting more productively is the lever

Appendix

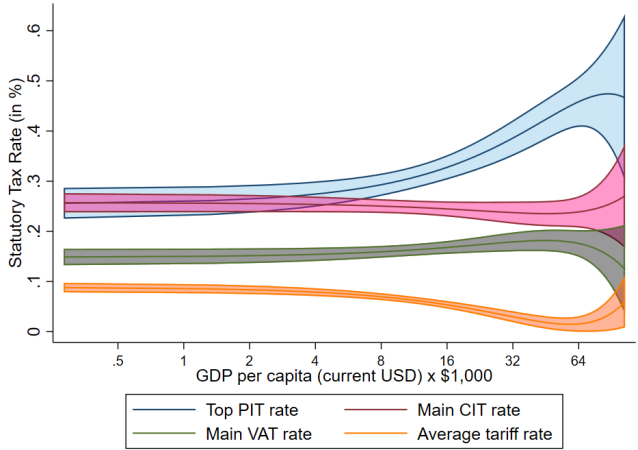
Descriptive Cross-Country Regressions: T/X and GDP per Capita

Outcome	(1) Log T/X	(2) Log T/X	(3) Log T/X	(4) Log T/X	(5) Log T/X	(6) Log T/X	(7) Log T/X
Log GDP p.c.	0.344***	0.773***	0.172*	0.334***	0.347***	0.358***	0.309***
	(0.048)	(0.166)	(0.088)	(0.070)	(0.048)	(0.048)	(0.058)
N	98	20	78	97	95	88	71
R^2	0.346	0.544	0.047	0.367	0.353	0.383	0.291
Modif. to column (1)		Only low, lower-middle inc. countries	Only upper-middle, high inc. countries	Cont. for PIT, VAT, CIT, tariff	Excl. if no VAT or sales tax	Excl. if no customs	Excl. if resp. for soc. sec.

Robust standard errors. Columns (2)–(3) split the sample at GDP per capita of $4,200. Dropping the two countries whose tax authority also hosts the population registry (Sweden, Norway) leaves the slope essentially unchanged at 0.336. * p<0.10 ** p<0.05 *** p<0.01.

Statutory Tax Rates Back to Main

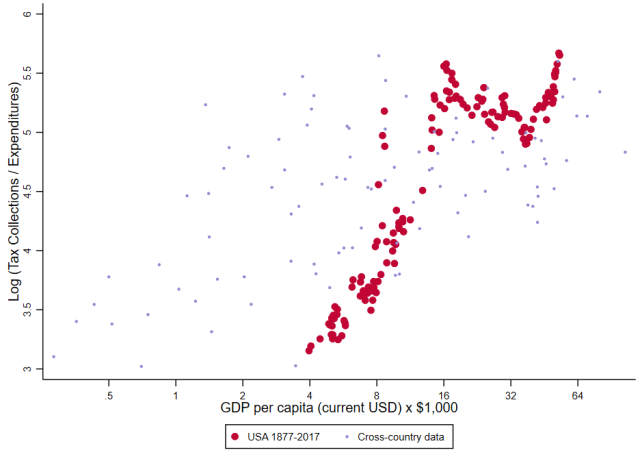
Statutory Tax Rates by GDP per Capita



Predicted average statutory rates: PIT, CIT, VAT, tariffs. Source: Bachas et al. (2024).

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The United States over the Long Run



US federal T/X (IRS + customs, digitized annual reports) overlaid on the modern cross-section.

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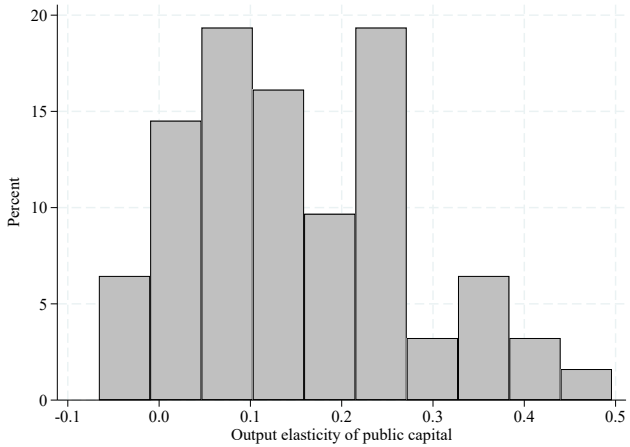
Long-Run Panel: 16 Countries, US States, Indian States

Outcome:	(1)	(2)	(3)	(4)	(5)
	Long-run change in Log T/X				
Long-run change in Log GDP p.c.	0.572**	0.614**	0.566**	1.358***	0.607***
	(0.218)	(0.240)	(0.225)	(0.489)	(0.210)
N (long-run changes)	32	28	26	50	60
Sample	Historical panel of countries			US states (1993–2017)	Indian states (1950–2017)
Countries / states	16	14	13	50	30

Each series is split into two approximately equal segments (average ≈27 years); the long-run change in log T/X is regressed on the long-run change in log GDP p.c., with segment fixed effects, so time-invariant characteristics are differenced out. Column (2) drops country-years where customs are not included in T/X; column (3) drops countries whose tax authority collects social security; column (4) uses a single long-run change per US state; column (5) uses two segments per Indian state (1950–1983, 1984–2017). SEs clustered by country/state (column 4: robust). Annual panel with country (state) and year fixed effects: countries 0.150 (0.216); Indian states 0.024 (0.096); US states 1.017 (0.098). * p<0.10 ** p<0.05 *** p<0.01.

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Output Elasticity of Public Capital: Meta-Analysis



Histogram of estimates from Bom and Ligthart (2014); 93% from advanced economies.

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Robustness: Targeting T/X or Tax-to-GDP Alone

$A_C \times 1.5$		Both (baseline)	T/X only	T/Y only
Low income	Δ GDP (%)	18.8	18.9	13.8
	Δ Consumption (%)	18.3	18.7	6.2
	Δ Tax rate (pp)	0.66	0.23	9.39
Middle income	Δ GDP (%)	14.7	17.1	11.4
	Δ Consumption (%)	9.7	15.0	5.4
	Δ Tax rate (pp)	8.03	3.64	9.45
High income	Δ GDP (%)	8.6	12.6	7.6
	Δ Consumption (%)	3.9	8.4	3.2
	Δ Tax rate (pp)	8.66	7.71	8.10

Each column recalibrates the model to both fiscal moments or to one of them; L_C is chosen optimally before and after the shock. Unweighted means across 10 low-, 46 middle-, and 42 high-income countries.

- Main result unchanged: large GDP gains, largest in low-income countries
- Targeting T/Y alone matches poor countries' higher tax-to-GDP ratios: a higher starting tax rate, so taxes absorb more of the gain

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Tax-Authority Activities and A_C: Individual Measures

Bivariate regression of A_C on:	Coefficient		R^2
Third-party data and digitization			
Computerized third-party data integration	0.449***	(0.083)	0.149
Pre-filing of tax returns	1.021***	(0.173)	0.256
Robotic process automation	0.952***	(0.183)	0.226
Data-science analytics	0.944***	(0.188)	0.123
Whole-of-government identification	0.676***	(0.185)	0.108
Artificial intelligence	0.791***	(0.184)	0.149
Enforcement			
Garnishment of wages and other payments	1.041***	(0.236)	0.082
Offset arrears against other domestic taxes	0.899***	(0.318)	0.032
Offset arrears via foreign tax authorities	0.460**	(0.216)	0.047
Operational processes			
Automated compliance checks	0.544***	(0.112)	0.217
Randomized audits for research and risk profiling	0.277*	(0.156)	0.031

Each row is a separate bivariate regression, $N = 97$, robust s.e. in parentheses; A_C standardized; continuous regressors standardized. Source: ISORA.

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Validating A_C: Robustness and Other ISORA Dimensions

- **Alternative data source** — OECD Inventory of Tax Technology Initiatives ($N = 59$):
  - Combined index: 0.445^{***} (0.137); 0.294^{**} (0.131) with D_i controls; -0.039 (0.076) when the outcome is A_M
  - Oyebola–Gabe dimensions: third-party data and detection 0.391^{***}; identification capacity 0.434^{**}; collection 0.110

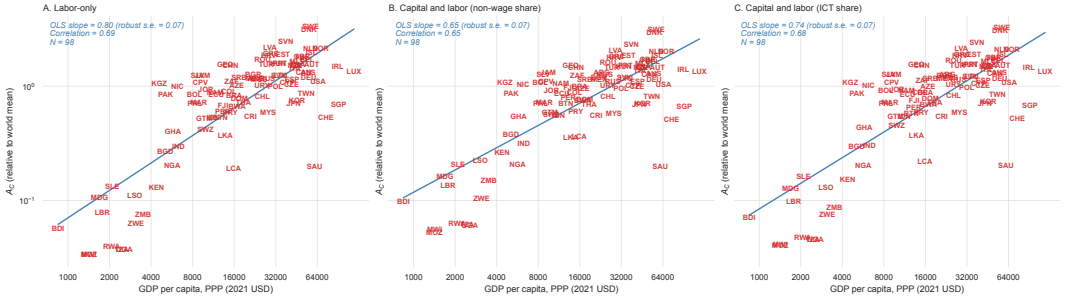
- **Imputing missing** D_i restores the full sample ($N = 97$): combined index 0.336^{***} (0.118); data 0.239^{**}, enforcement 0.182^{**}, operations 0.199^{**}; A_M placebo -0.035 (0.067)

- **Other ISORA dimensions** (all $N = 96$, no controls): taxpayer services 0.182; human resources 0.009; institutional autonomy 0.040; large-taxpayer unit -0.027 — none significant
  - Limited effects or limited power (cf. Basri, Felix, Hanna, and Olken 2021 on large-taxpayer offices)

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Robustness: Capital in Tax Collection

- $\tau = A_C K_C^{\eta\psi} L_C^{\eta(1-\psi)} = \left[A_C (K_C/L_C)^{\eta\psi}\right] L_C^{\eta}$: a labor-only model folds capital per collector into A_C
- $X = wL_C + \delta_C K_C$; set $\delta_C K_C / X$ to the income-group mean non-wage (or ICT) share, recalibrate all countries
- Share of operating expenditure, low / middle / high income: non-wage 26.8 / 33.3 / 27.4%; ICT 11.5 / 8.5 / 15.1%

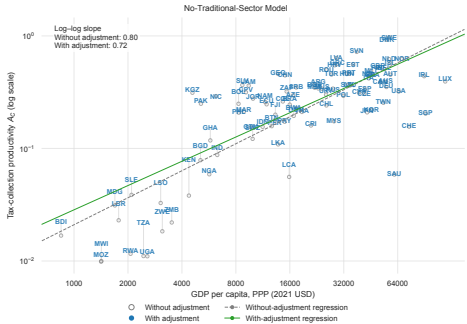


Slope of $\log A_C$ on log GDP p.c.: 0.80 (labor only), 0.65 (non-wage share), 0.74 (ICT share). Low-income ICT share observed for Malawi only.

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Robustness: Public-Sector Wage Premia

- Baseline pays collectors the private-sector wage $\Rightarrow$ a low T/X loads entirely onto a low A_C
- Let $\pi_i = w_{C,i}/w_{M,i}$ (Abdallah, Coady, and Jirasavetakul 2023; direct for 67 of 98 countries, rest imputed): given T/X and L_C, $\log A_C$ shifts up by $\approx \log \pi_i$, most in poor countries
- Log–log slope: $0.80 \rightarrow 0.72$ $\Rightarrow$ wage premia account for about a tenth of the gradient



Baseline model (no traditional sector). The adjustment also enters the public-capital budget:

$$\delta_G K_G = (1-\xi)\tau w_M (L_M + \pi L_C) - \pi w_M L_C.$$

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Steady State

Given (L_M, K), public capital and consumption follow from:

$$K_G = \left[\frac{(1-\mu) A_M \theta^{\zeta} K^{\mu} L_M^{-\mu} \left[(1-\xi)\tau(L_M + L_C) - L_C \right]}{\delta_G} \right]^{\frac{1}{1-\zeta}},$$

$$w = (1-\mu) A_M (\theta K_G)^{\zeta} K^{\mu} L_M^{-\mu}, \qquad C = \left(\frac{(1-\tau)w}{\alpha (L_M + L_C)^{1/\varepsilon}} \right)^{1/\gamma}.$$

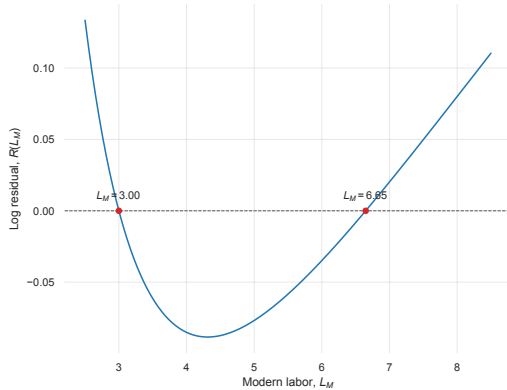
Steady state characterized by two equations in two unknowns (L_M, K):

$$\mathcal{R}_1(L_M, K) \equiv 1 - \beta \left[1 + \mu A_M (\theta K_G)^{\zeta} K^{\mu-1} L_M^{1-\mu} - \delta \right] = 0,$$
$$\mathcal{R}_2(L_M, K) \equiv C + \delta K + \delta_G K_G - A_M (\theta K_G)^{\zeta} K^{\mu} L_M^{1-\mu} = 0.$$

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Existence, Uniqueness, and Multiple Steady States

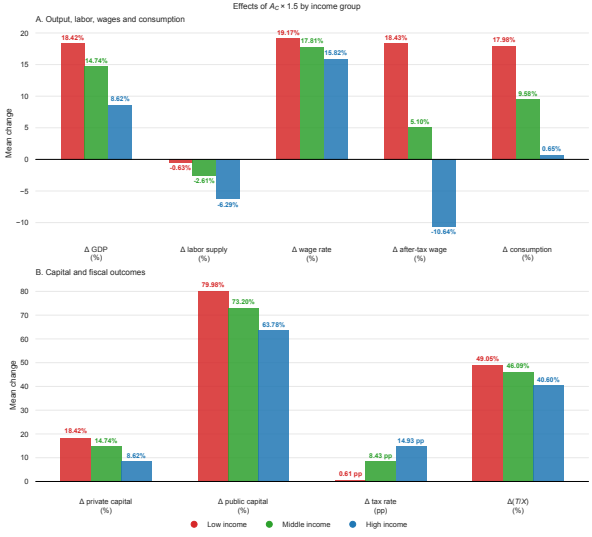
- **Proposition**: if $\gamma > 1$ and $\mu + \zeta < 1$, a unique steady state exists
- The steady state reduces to one equation in L_M:
  - log marginal disutility of work: increasing in L_M
  - log marginal utility of after-tax earnings: decreasing when $\gamma > 1$ and $\mu + \zeta < 1$
- Public capital raises taxable income, hence revenue and public capital again: outside these conditions this feedback can support **multiple steady states**
- Example: $\gamma = 2$, $\mu = 0.5$, $\zeta = 1 \Rightarrow$ steady states at $L_M = 3$ and $L_M \approx 6.65$



Residual $\mathcal{R}(L_M)$: log marginal disutility of work minus log marginal utility of after-tax earnings

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Raising A_C by 50 Percent: Capital Stocks and Tax Rates



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Endogenous Collection with Inelastic Labor: Why η Matters

- With inelastic labor the government maximizes $C = (1-\varphi)Y$; its first-order condition in L_C:

$$\eta\left(\Lambda_0 - W_c\right) \;=\; \frac{\lambda_C}{1-\zeta}, \qquad \Lambda_0 = \frac{\zeta}{1-\zeta}, \quad W_c = \frac{\varphi}{1-\varphi}$$

  η log points of tax rate, each worth $\Lambda_0 - W_c$ in consumption = output cost of the collector

	$\eta = 0.05$	$\eta = 0.20$	$\eta = 0.50$	$\eta = 1$
Elasticity of output to tax productivity				
Low income	0.250	0.247	0.148	0.013
Middle income	0.234	0.136	0.027	0.007
High income	0.209	0.094	0.022	0.007
Optimal public investment rate $\varphi^\star$, against $\zeta = 0.20$				
Low income	0.027	0.050	0.150	0.195
Middle income	0.141	0.172	0.193	0.198
High income	0.168	0.184	0.194	0.198

- As η rises, $\varphi^\star \to \zeta$ (the Barro optimum): a better A_C buys a smaller payroll, not more output

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Ramsey Extension: The Government's Problem

- Household supplies $L = L_{M,t} + L_{C,t}$ inelastically and solves $\max \sum_t \beta^t u(C_t)$ s.t.

$$C_t + K_{t+1} + b_{t+1} = (1 - \omega_t)(1 + r_t - \delta)K_t + R_t b_t + (1 - \tau_t) w_t L + TR_t$$

- Public-investment account, as in the baseline: $K_{G,t+1} = (1 - \delta_G)K_{G,t} + \Phi(L_{C,t})\, w_t$, $\Phi(L_C) = (1 - \xi)\tau(L_C)L - L_C$

- General account: borrows and sets the capital wedge (incl. a levy on initial wealth), $R_t b_t = b_{t+1} + \omega_t(1 + r_t - \delta)K_t$ — but cannot fund public investment

- Implementability is slack (initial-wealth levy) $\Rightarrow$ primal problem: $\max \sum_t \beta^t u(C_t)$ subject to

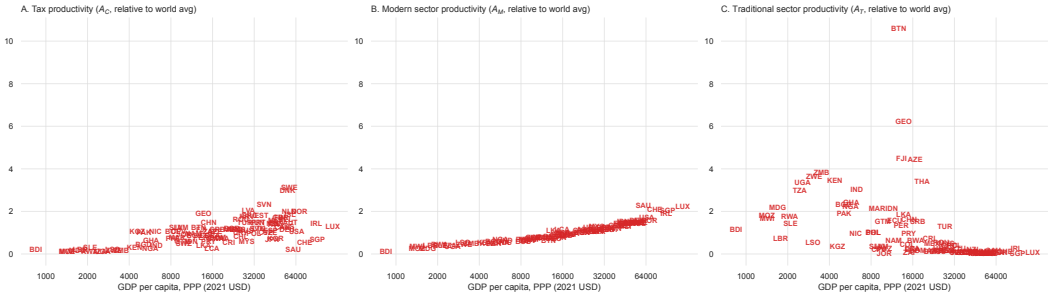
$$C_t + K_{t+1} + K_{G,t+1} = F(K_t, K_{G,t}, L - L_{C,t}) + (1 - \delta)K_t + (1 - \delta_G)K_{G,t}$$

  and the law of motion of K_G (Lucas and Stokey 1983; Chari and Kehoe 1999)

- Steady state: $\beta R = 1$; the government also subsidizes private capital, which raises the tax base

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Traditional-Sector Extension: Calibrated Productivities



A_T calibrated to each country's agricultural employment share; labor disutility weights $\alpha_M = \alpha_T = 1$.

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Traditional-Sector Extension: Policy Counterfactuals

Shock	Income group	GDP	Consumption	K_G	After-tax wage	T/X	$\Delta\tau$ (pp)
$A_C \times 1.5$	Low income	1.06	1.05	2.02	1.22	1.61	1.32
	Middle income	1.13	1.07	1.73	1.03	1.46	9.57
	High income	1.08	1.00	1.63	0.89	1.40	15.14
$A_M \times 1.5$	Low income	1.50	1.42	3.92	2.74	1.40	0.00
	Middle income	2.22	2.19	2.38	2.37	1.00	0.00
	High income	2.28	2.28	2.28	2.34	0.97	0.00
$L_C \times 1.5$	Low income	1.00	1.00	1.00	1.00	0.68	0.05
	Middle income	1.00	1.00	1.01	1.00	0.68	0.39
	High income	1.00	1.00	1.01	0.99	0.68	0.66
$\theta = 0.84$	Low income	1.06	1.05	1.30	1.21	1.07	0.00
	Middle income	1.05	1.05	1.06	1.06	1.00	0.00
	High income	1.00	1.00	1.00	1.00	1.00	0.00

New steady state relative to baseline; $\Delta\tau$ in percentage points; unweighted means within World Bank income groups. Denmark and Sweden are excluded from $A_C \times 1.5$ (implied $\tau \geq 1$).

Back Hulten

Traditional-Sector Extension: $A_C \times 1.5$ vs. Hulten

Prediction method (low-income countries)	Δ GDP (%)
Model	6.22
Hulten (wage-bill share: $0.5 \times w_M L_C / Y$)	0.01
Hulten (tax share: $0.5 \times T/Y$)	0.31
Public capital channel ($\zeta \times \%\Delta K_G \times Y_M/\text{GDP}$)	5.67

- Public-capital row scales the modern-sector contribution $\zeta \times \%\Delta K_G$ by the baseline modern-sector GDP share Y_M/GDP, to make it comparable to aggregate GDP

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Robustness: Targeting Unmechanized Agriculture

- Alternative target: agricultural employment $\times$ (1 $-$ mechanization rate), with mechanization linear in log GDP p.c. (31.9% for Malawi/Nigeria/Tanzania/Uganda average; 100% for richest)

Shock	Income group	GDP	Consumption	K_G	After-tax wage	T/X	$\Delta\tau$ (pp)
$A_C \times 1.5$	Low income	1.13	1.11	1.89	1.18	1.54	2.14
	Middle income	1.14	1.09	1.73	1.04	1.46	9.05
	High income	1.08	1.00	1.64	0.89	1.40	15.09
$A_M \times 1.5$	Low income	1.95	1.87	2.92	2.51	1.15	0.00
	Middle income	2.27	2.27	2.28	2.34	0.98	0.00
	High income	2.28	2.28	2.28	2.34	0.97	0.00
$L_C \times 1.5$	Low income	1.00	1.00	1.01	1.00	0.68	0.09
	Middle income	1.00	1.00	1.01	1.00	0.68	0.37
	High income	1.00	1.00	1.01	0.99	0.68	0.66
$\theta = 0.84$	Low income	1.12	1.11	1.23	1.19	1.03	0.00
	Middle income	1.05	1.05	1.05	1.05	1.00	0.00
	High income	1.00	1.00	1.00	1.00	1.00	0.00

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Optimal Collection Labor by Income Group

Income group	L_C/L_C^*	L_C/L (%)	Δ Utility	ΔC (%)	Δ GDP (%)	ΔK_G (%)
Low income	0.78	0.03	0.00	0.12	0.12	0.48
Middle income	0.53	0.16	0.01	0.38	0.64	3.31
High income	2.37	0.25	0.01	0.76	0.38	0.02

- L_C^*: welfare-maximizing steady-state choice of collection labor, holding all calibrated parameters fixed; Δ columns report gains from moving to L_C^* (unweighted means of country-level changes)
- L_C/L: calibrated collection employment share — well under 1% of the workforce everywhere

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